

Measuring Drift in Therapeutic AI: A Stability-Based Evaluation of Sovereign LLMs in Nightmare Therapy

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Abstract

[TODO: Abstract: RQ (run-to-run consistency of LLM clinical reasoning), three-level framework (plan consistency / response similarity / plan-response alignment), 11 models $\times$ 6 vignettes $\times$ 5 temperatures $\times$ 20 trials = 6,600 trials, headline findings incl. temperature effect and extensibility validation via post-hoc Mistral Medium 3.5 addition.]

1 Introduction

Nightmare disorder affects roughly 4% of adults (Sandman et al., 2013; Li et al., 2010), and Imagery Rehearsal Therapy (IRT) is its best-evidenced treatment (Krakow et al., 2001; Morgenthaler et al., 2018). Access, however, is scarce, motivating LLM-based delivery of structured protocols (Stade et al., 2024; Heinz et al., 2025). A therapeutic LLM must do more than produce plausible text: within a structured protocol it must make *consistent clinical decisions* — a patient re-entering the rescripting phase should not receive a confrontation strategy on one run and an avoidance-incompatible safety strategy on the next, purely by chance of sampling.

Existing evaluations of clinical LLMs focus on quality, safety, or efficacy of single outputs (Zhang et al., 2024; Chiu et al., 2024); run-to-run *stability* at fixed input is rarely measured, despite growing evidence that sampling variance materially affects LLM behavior (Renze, 2024; Novikova et al., 2025). [TODO: Sharpen the gap statement against 2–3 closest works.]

This paper. We frame run-to-run consistency of clinical reasoning as a measurable property and evaluate it in the rescripting phase of IRT. Models declare their therapeutic strategy in a structured <plan> block before responding, enabling measurement at three levels: (1) *plan consistency* — pairwise Jaccard similarity of declared strategy sets; (2) *response similarity* — pairwise BERTScore of the therapeutic responses; (3) *plan-response alignment* — an LLM judge verifying that responses implement the declared strategies.

Contributions.

1. A three-level, protocol-grounded stability evaluation framework for therapeutic LLMs, applied to the rescripting core of IRT with frozen dialogue histories as deterministic evaluation entry points.
2. An empirical study of 11 models $\times$ 6 patient vignettes $\times$ 5 temperatures $\times$ 20 trials (6,600 trials), quantifying decision-level and response-level stability and their temperature dependence. [TODO: one-sentence headline finding once results are final]

3. An extensibility validation: Mistral Medium 3.5 was added post hoc, four months after the main runs, on identical frozen histories with a single configuration entry — testing both the pipeline’s extensibility claim and the robustness of the findings to model-set enlargement.

2 Related Work

IRT and digital delivery. Imagery Rehearsal Therapy is the first-line psychological treatment for nightmare disorder (Krakow et al., 1995, 2001; Morgenthaler et al., 2018), with rescripting as its cognitive core (Albanese et al., 2022). Digital and conversational delivery of structured therapy protocols has shown feasibility (Fitzpatrick et al., 2017; Heinz et al., 2025; Wu et al., 2026). [TODO: 1–2 sentences tying reDreamAI / prior efficacy study (Llama 3.3 70B continuity).]

Evaluating clinical LLMs. Existing frameworks target competence and safety of single responses (Zhang et al., 2024; Chiu et al., 2024; Iftikhar et al., 2025; Stade et al., 2024); protocol fidelity instruments from clinical psychology (Mowbray et al., 2003; Bellg et al., 2004; Kohrt et al., 2015) inspire our ternary alignment rubric. Our focus differs: not *whether one response is good*, but *whether repeated responses agree*.

LLM output consistency. Sampling temperature and stochastic decoding measurably change task behavior (Renze, 2024); recent work documents self-inconsistency across repeated runs (Novikova et al., 2025; Xu et al., 2025; Yuan et al., 2025). [TODO: Position against these: none measure decision-level consistency in a clinical protocol.]

Chain-of-thought and declared intent. CoT prompting (Wei et al., 2022) structures generation, but CoT explanations are often unfaithful to internal computation (Turpin et al., 2023; Lanham et al., 2023). We therefore frame the plan block as declared intent (Section 3.2). LLM-as-judge protocols (Zheng et al., 2023) underpin Method 3, with cross-family judging to mitigate self-preference bias.

3 Method

3.1 Two-Stack Architecture

The evaluation system separates dialogue creation from stability measurement (Figure 1). A *generation stack* produces synthetic therapy sessions through a three-agent loop: a patient agent simulates responses based on a BDI-profiled vignette, a router agent classifies the current stage of the Imagery Rehearsal Therapy (IRT) protocol from the conversation history, and a therapist agent generates stage-appropriate responses. The loop traverses the five-stage IRT protocol (recording, rewriting, summary, rehearsal, final) and exports the dialogue as a *frozen history*, together with prefixes sliced at rewriting-turn boundaries.

An *evaluation stack* then measures stability on these frozen histories. The router is bypassed—the rescripting prompt is injected directly, since the protocol stage is known a priori—which isolates the therapist model from stage-classification error. Every trial receives byte-identical context: the frozen history sliced after the second rewriting exchange (slice 2), followed by the same rescripting instruction.

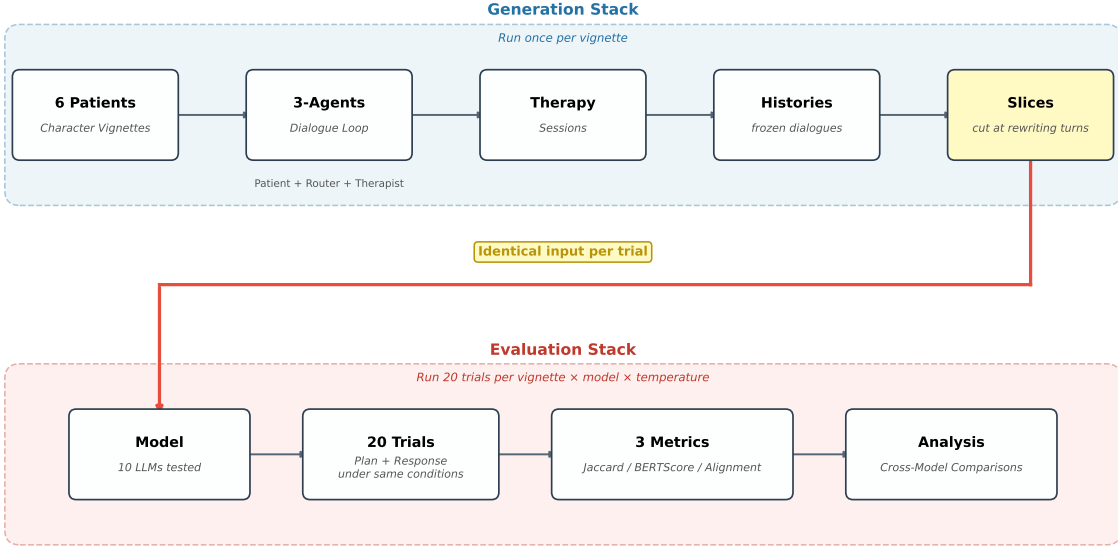


Figure 1: Pipeline overview: vignettes feed the generation stack; frozen histories feed the evaluation stack; each experiment run produces per-trial outputs and aggregated metrics.

3.2 The Plan Mechanism

Each trial uses *fused* chain-of-thought generation: the model first emits a plan block, `<plan>strategy_1 / strategy_2</plan>`, declaring one or two therapeutic strategies from a fixed taxonomy, then produces its therapeutic response in the same call, so the plan tokens condition the response tokens.

The plan block is a *declared intent* mechanism, not a claim about internal reasoning. The chain-of-thought faithfulness debate (Turpin et al., 2023; Lanham et al., 2023) is acknowledged but deliberately sidestepped: whether or not the declaration reflects internal computation, variable declarations predict variable clinical responses, and the declaration is what a clinical overseer would review.

3.3 Strategy Taxonomy

Declarations are constrained to six categories describing concrete therapeutic *mechanisms* (not goals): **confrontation** (external action toward the threat), **self_empowerment** (internal transformation), **safety** (environment modification), **cognitive_reframe** (meaning change), **social_support** (adding allies), and **sensory_modulation** (sensory and calming details). The taxonomy was refined over four iterations to resolve a severe distribution skew in which a single broad “empowerment” category absorbed 87.7% of all picks; the final six categories are mechanism-level and mutually discriminable (Appendix A).

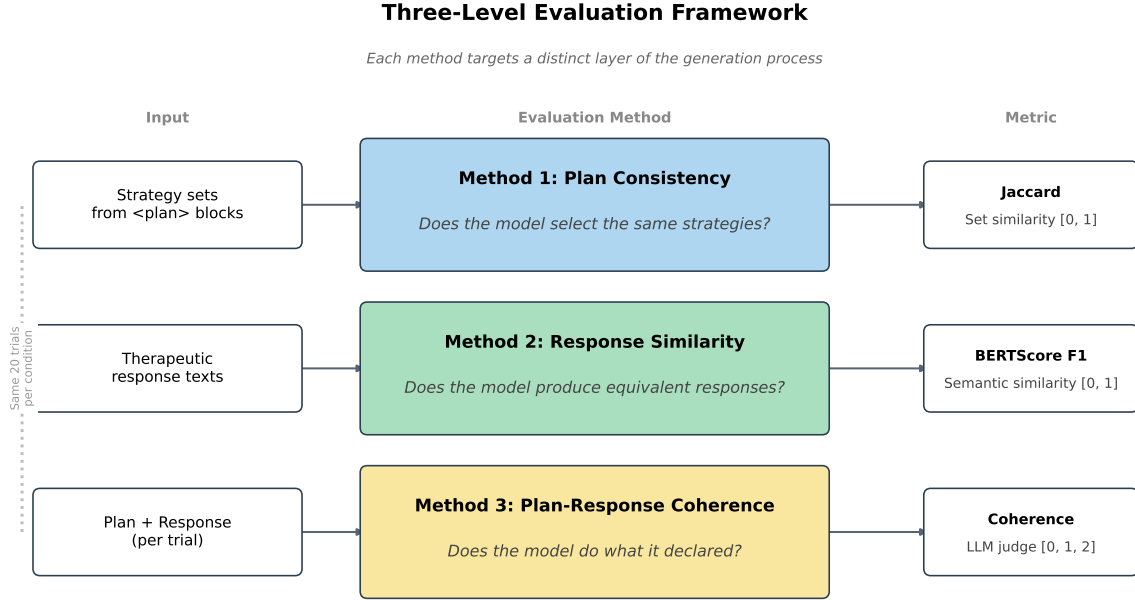


Figure 2: The three-level measurement framework: each trial’s output feeds plan consistency (Method 1), response similarity (Method 2), and plan–response alignment (Method 3).

3.4 Three Evaluation Metrics

Each experiment run consists of $N = 20$ independent trials of one model on one vignette at one temperature. Three metrics target distinct layers of consistency (Figure 2):

Method 1: Plan consistency. Do stochastic runs produce the same therapeutic *decisions*? We compute the mean pairwise Jaccard similarity over the declared strategy sets across all $\binom{20}{2} = 190$ trial pairs. A plan *validity rate* (fraction of trials whose plan block parses to taxonomy categories) serves as an upstream quality gate.

Method 2: Response similarity. Do stochastic runs produce therapeutically equivalent *responses*? We compute mean pairwise BERTScore F1 (Zhang et al., 2020) over the response texts (plan blocks excluded) across the same 190 pairs, using DeBERTa-XLarge-MNLI (He et al., 2021) as the embedding model, which ranks first of 130+ models on WMT16 human correlation ($r = 0.778$); its NLI fine-tuning aligns with the semantic-equivalence judgment the metric requires.

Method 3: Plan–response alignment. Does the model *do what it says*? An LLM judge (Gemini 3 Flash at $T=1.0$; suspect trials re-judged by Gemini 3.1 Pro) scores each declared strategy on a ternary scale (0 = absent, 1 = partial, 2 = implemented), normalized to $[0, 1]$ per trial. Bias mitigations: cross-family judging (the judge’s model family does not appear among evaluation targets), chain-of-thought justification, and a transparent rubric (Appendix C). Judge trials that fail (timeout or API error) or are unjudgeable (invalid plan) are treated as missing data and excluded from the alignment mean rather than scored zero; invalid plans are already penalized by the validity rate.

Model	Size	Class	License	Collected
Mistral Small 3.2	24B dense	small	Apache 2.0	2026-03
Mistral Small 4	119B MoE (6.5B act.)	large MoE	Apache 2.0	2026-03
Mistral Medium 3.5	128B dense	large dense	modified MIT	2026-07
Mistral Large 3	675B MoE (41B act.)	large MoE	Apache 2.0	2026-03
Qwen 3.5 27B	27B dense	small	Apache 2.0	2026-03
Qwen 3.5 122B	122B MoE (10B act.)	large MoE	Apache 2.0	2026-03
Qwen 3.5 397B	397B MoE (17B act.)	large MoE	Apache 2.0	2026-03
OLMo 3.1 32B	32B dense	small	Apache 2.0 (full provenance)	2026-03
Llama 3.3 70B	70B dense	mid	Llama license	2026-03
GPT-5.4	undisclosed	proprietary	—	2026-03
Claude Sonnet 4.6	undisclosed	proprietary	—	2026-03

Table 1: Evaluation targets (therapist role). All models run in non-thinking mode via OpenRouter. Supporting roles are fixed across all runs: patient = Dolphin Mistral Venice 24B, router = Llama 3.3 70B, judge = Gemini 3 Flash with Gemini 3.1 Pro fallback (no family overlap with any evaluation target).

Alignment primarily serves as a validation check: scores near the ceiling confirm that models implement what they declare, validating Jaccard as a measure of genuine decision stability rather than label noise.

4 Experimental Setup

4.1 Models

We evaluate 11 models in the therapist role (Table 1), all in non-thinking mode via OpenRouter. The primary subject is Mistral Large 3, the EU-sovereign Apache-2.0 flagship; sovereignty motivates this choice but imposes no constraint on the comparison set. The set spans a size-class ladder from 24–32B dense models through 70B dense and 119–675B MoE up to two proprietary ceilings, with three scales of Qwen 3.5 probing within-family scaling and OLMo 3.1 providing full training-data provenance.

Mistral Medium 3.5 (128B dense, open weights under a modified MIT license, released 2026-04-30) was added *post hoc* in July 2026, four months after the main data collection, using the identical frozen histories, prompts, and analysis pipeline. This addition serves two purposes: it validates that the framework extends to a new model with a single configuration entry, and it tests whether the main findings are robust to enlarging the model set (Section 6.1).

4.2 Conditions

The full design crosses 11 models $\times$ 6 vignettes (anxious, avoidant, cooperative, resistant, skeptic, trauma) $\times$ 5 temperatures ($T \in \{0.0, 0.075, 0.15, 0.3, 0.6\}$) $\times$ 20 trials at slice 2, i.e. 600 trials per model and 6,600 trials in total. Ten models were collected 2026-03-21 to 2026-03-24; Mistral Medium 3.5 on 2026-07-09 (a `run_date` column in the released data marks this).

4.3 Statistical Analysis

All tests are exploratory. Temperature effects are assessed with Spearman rank correlations (pooled and per model) and Kruskal–Wallis H across temperature groups; model and vignette differences

with Kruskal–Wallis H plus pairwise Mann–Whitney U under Bonferroni correction ($\binom{11}{2} = 55$ comparisons); cross-metric relations with Spearman correlations. Effect sizes (η^2) are reported alongside p -values. No absolute threshold for “stable enough” exists in the literature; results are interpreted comparatively.

4.4 Reproducibility

Model endpoints are pinned by OpenRouter model IDs; exact IDs, run dates, prompts, frozen histories, and per-trial outputs are released with the code. Two caveats: OpenRouter routes requests to third-party inference providers whose serving stacks (e.g. quantization) may differ, and the judge endpoints are versioned previews; both are discussed in Section 6.2.

5 Results

[TODO: All numbers below from `stats/data/experiment_test_results.json` and `experiment_descriptives.js` after the 11-model aggregation. Structure mirrors the thesis results chapter.]

5.1 Plan Validity and Strategy Distribution

[TODO: validity rates per model (fig_5_1 / fig_A1); strategy distribution across taxonomy.]

5.2 Plan Consistency (Jaccard)

[TODO: fig_5_2 by family panels + fig_5_2b modal-set agreement; temperature trend; model differences w/ Kruskal–Wallis + Bonferroni pairwise.]

5.3 Response Similarity (BERTScore)

[TODO: fig_5_3; compare decision-level vs response-level stability.]

5.4 Vignette Effects

[TODO: fig_5_4.]

5.5 Cross-Metric Relations

[TODO: fig_5_5; Spearman correlations between the three methods.]

5.6 Plan–Response Alignment

[TODO: fig_5_6; alignment as validation check; report `n_judged` fraction, pro-fallback fraction, and the missing-data rule for judge failures.]

5.7 Robustness to Model-Set Enlargement: Mistral Medium 3.5

The post-hoc addition lands at the top of the stability ranking: Mistral Medium 3.5 attains a perfect median Jaccard of 1.000, the highest mean BERTScore F1 in the set (0.816), perfect plan validity (1.000), and mean alignment of 0.961 across its 30 conditions. Its Jaccard temperature profile stays at or near ceiling across all five temperatures.

Enlarging the pairwise family from $\binom{10}{2} = 45$ to $\binom{11}{2} = 55$ comparisons tightens the Bonferroni correction; exactly two marginal pairs among the original ten models lost significance (Jaccard: Llama 3.3 70B vs. Qwen 3.5 122B, $p_{\text{bonf}} 0.046 \rightarrow 0.056$; alignment: GPT-5.4 vs. Mistral Large 3, $0.045 \rightarrow 0.055$). No pairwise conclusion reversed direction, no pair became newly significant among the original ten, and the BERTScore pairwise structure is unchanged. All substantive findings of the original study hold under model-set enlargement.

6 Discussion

[TODO: Interpret: what do the stability levels mean for clinical deployment; temperature guidance; EU-sovereign subject vs proprietary ceiling verdict.]

6.1 Robustness and Extensibility

Adding Mistral Medium 3.5 four months after the main data collection required a single configuration entry (model ID, temperature, reasoning mode) and one command; the 600-trial collection completed in roughly 8.5 hours of unattended wall-clock time on the identical frozen histories. No analysis code changed: aggregation, tests, and figures picked up the eleventh model automatically. The framework’s extensibility is thus a demonstration rather than a claim, and the demonstration doubles as a robustness check — Section 5.7 shows all substantive findings survive the enlargement, with only two marginal Bonferroni boundary crossings. [TODO: Optionally add API cost figure.]

6.2 Limitations

- **Synthetic patients.** Vignette-driven patient simulation, not clinical interactions.
- **Serving-stack variance.** OpenRouter routes to third-party providers; quantization and batching may differ between requests and between March and July collections.
- **Judge versioning.** Alignment scores depend on versioned Gemini preview endpoints; the judge for the July runs may differ in revision from March. Judge reliability is itself an open question (Zheng et al., 2023).
- **Single protocol phase.** Rescripting only; stability elsewhere in IRT is unmeasured.
- **Exploratory statistics.** 30 runs per model; Bonferroni-corrected pairwise tests are conservative.

7 Conclusion

[TODO: Half page: framework contribution, headline empirical findings, extensibility demonstration, outlook (human validation of judge, other protocol phases, real-world pilots).]

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A Strategy Taxonomy

[TODO: Taxonomy card: six categories with definitions and example phrases; brief evolution note (8 → 6 → 7 → 6, skew collapse figure).]

B Prompts

[TODO: Rescripting prompt, fused-generation instruction, vignette example.]

C Judge Rubric

[TODO: Judge system prompt, ternary rubric, example judgment with CoT justification; missing-data rule for judge failures.]

D Per-Condition Results

[TODO: Full 11-model $\times$ 5-temperature tables for all three metrics.]